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1. **Functional Requirements**

This section describes actions that BatchBite shall do. The requirements are categorized based on modules and are assigned a priority level (High, Medium, and Low) along with the sprint during which they will be implemented. Requirements for Sprint 1 make up the primary core requirements; others are added now to incorporate them into the design.

1.1 Accounts and Access

ID	Requirement	Priority	Sprint
FR-1	The system will provide role-based account options for Customers, Business Owners, Kitchen Staff, and Administrators.	High	1
FR-2	The system will provide options for secure registration and login, and will limit access to all screens and functions based on the role permitted to access them.	High	1
FR-3	The system will provide Administrators with the ability to manage user accounts and set or modify user roles.	Medium	1

1.2 Products and Production Batches

ID	Requirement	Priority	Sprint
FR-4	The system shall provide Business Owners with the ability to create and	High	1

	modify listings of their products with pictures, descriptions, variants, ingredients information, pricing, and available quantity.		
FR-5	The system shall provide Business Owners with the ability to create batches for product manufacturing with an order deadline, maximum quantity, and pick-up date.	High	1
FR-6	The system shall automatically close orders for a particular batch upon reaching the deadline or maximum quantity.	High	1

1.3 Cart, Checkout, and Stock Reservation

ID	Requirement	Priority	Sprint
FR-7	The system shall allow Customers to browse open batches and place products in their cart.	High	1
FR-8	The system shall show the order summary (Items, quantities, total, pickup slot) before the customer confirms the checkout.	High	1

FR-9	The system shall confirm a payment through a simulated payment confirmation.	High	1
FR-10	The system shall reserve the stock when the checkout occurs by holding it, in a way that the stock will not be accessible to another Customer while the hold is active	High	1
FR-11	The system shall cancel a reservation of the stock and the capacity when a hold expires or when an order is cancelled.	High	1

1.4 Pickup Slots

ID	Requirement	Priority	Sprint
FR-12	The system shall enable the Business Owners to set pickup timeslots with maximum orders per time slot.	High	1
FR-13	The system shall enable Customers to choose a suitable pickup time slot and prevent them from selecting time slots that are full.	High	1

1.5 Order Tracking and Production Workflow

ID	Requirement	Priority	Sprint

FR-14	The system shall monitor each order throughout the process of Confirmed, Preparing, Ready, and Collected.	High	1
FR-15	The system shall only permit valid status changes (from Confirmed to Preparing to Ready to Collected) and will prevent invalid status changes such as marking an order as Collected before its status becomes Ready.	High	1
FR-16	The system shall enable customers to access their order history and status of each order.	High	1
FR-17	The system shall provide a Business Owner Dashboard to view orders, cancellation of orders, and progress of production.	High	1
FR-18	The system shall enable the Kitchen Staff to access orders for the current batch along with customization and packaging information.	High	1
FR-19	The system shall remove canceled	Medium	1

	orders from the active production board.		
FR-20	The system shall inform Customers about order confirmation, order ready for pick up, and delay notifications and maintain a record of all the notifications made accessible to the staff.	Medium	1
FR-21	The system shall consider two successive actions of a particular action as one action and shall not create duplicate records.	Medium	1

1.6 Order Changes and Cancellations

ID	Requirement	Priority	Sprint
FR-22	The system shall allow the Customers to modify quantity/variant prior to the cutoff provided only if there is sufficient stock and production capacity, recalculate the totals and the simulated refund but retain the same history of the order.	Medium	2
FR-23	The system shall maintain the recipe version and price version that were	Medium	2

	used for each confirmed order to ensure that future edits cannot affect the already confirmed orders.		

1.7 Ingredients, Collection, and Reporting

ID	Requirement	Priority	Sprint
FR-24	The system shall derive the required ingredients based on confirmed orders, scaling the recipes with the quantity of the order, compare them against available and reserved stock, and create a purchasing list for any shortfall.	Medium	2
FR-25	The system shall create a unique QR code for each order and only confirm the delivery of the order when it is Ready.	Medium	2
FR-26	The system shall organize the products into preparation batches on the production board retaining all customer-specific notes.	Low	2
FR-27	The system shall report the amount of orders confirmed/cancelled/produced/collected by batch and product.	Low	3

4. User Stories: Business Owner

The Business Owner is the owner of a small food business who schedules production batches, creates/manages orders, and tracks their progress. The stories below include acceptance criteria and functional requirements.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
BO-1	As a Business Owner, I	Product save occurs for all	FR-4	High	1

	would like to be able to create product listings with photos, variants, ingredients and pricing, in order to present the exact offerings to the customers.	fields and the product appears in the customer catalogue; errors appear for missing required fields.			
BO-2	As a Business Owner, I would like to be able to create production batches with deadlines, max quantities and collection dates, in order to avoid overselling.	Batch creation includes all three values; orders close automatically either at the set deadline or when the maximum quantity is reached.	FR-5, FR-6	High	1
BO-3	As a Business Owner, I would like to be able to create pickup slots with order limits, in order to avoid overcrowding during collection periods.	When the slot is full, the field is indicated as such and cannot be chosen by customers.	FR-12, FR-13	High	1

BO-4	As a Business Owner, I would like to be able to see all the orders in a particular batch in a dashboard, in order to get my confirmed demand.	Dashboard displays every order with information on customer, ordered items, current status and assigned pick-up slot along with the total amount of ordered products compared to maximum quantity.	FR-17	High	1
BO-5	As a Business Owner, I would like to be able to record any order cancellations, in order to release stock and capacity.	Cancelled order is not included in the active orders anymore; its quantity and pick-up slot go back into batch.	FR-11, FR-17, FR-19	Medium	1
BO-6	As a Business Owner, I would like to be able to update the production progress in order to keep order status updated for my customers.	Status update is reflected in customer view along with notification; invalid statuses are not accepted.	FR-14, FR-15, FR-20	Medium	1
BO-7	As a Business Owner, I	Order update can occur only	FR-22	Medium	2

	would like to be able to approve or deny the change requests of customers before the cut-off, in order to avoid exceeding my stock and capacity.	if the stock and capacity are sufficient enough; total amount and the refunded price are updated and previous order history is preserved.			
BO-8	As a Business Owner, I would like to be able to have ingredient requirements calculated based on confirmed orders.	Shopping list indicates required amount of ingredients compared to currently available amount.	FR-24	Medium	2
BO-9	As a Business Owner, I would like to have ordered, cancelled, produced and collected quantities in a report so that I can plan better future batches	Report can be filtered based on batch and product and compares planned with actual amount.	FR-27	Low	3

5. User Stories: Kitchen Staff

Kitchen Staff prepares and delivers orders. They need to be able to have a clear visibility of their tasks and to be able to track their progress and confirm delivery.

ID	User Story	Acceptance Criteria	Related FR	Priority	Sprint
KS-1	As Kitchen Staff, I need to see all the orders which are currently prepared, so I will know what should be prepared.	Order list contains the items, their variations, amount of products, and time slots for picking up of the chosen batch.	FR-18	High	1
KS-2	As Kitchen Staff, I need to see the customizations and packing details, so nothing special would be forgotten.	Customer's notes appear on each order in the production process.	FR-18	High	1
KS-3	As Kitchen Staff, I need to be able to set the order state as Preparing and Ready, so the progress could be tracked.	Only transition states of Confirmed to Preparing and Preparing to Ready are possible, and the customer receives notification about readiness of the order.	FR-14, FR-15, FR-20	High	1
KS-4	As Kitchen Staff, I need to be able to	Collected state is disabled for	FR-15	High	1

	confirm the delivery of an order only in case it is Ready, so wrong handovers will not happen.	orders which are not Ready and message appears about it.			
KS-5	As Kitchen Staff, I need cancelled orders to vanish from the production panel, so I will not prepare unnecessary orders.	Cancelled orders are invisible in the active board, but remain in the history of orders.	FR-19	Medium	1
KS-6	As Kitchen Staff, I need the ability to send the status update once even if I tapped twice, so I will not create duplicates.	Duplicate submission of the same update results in single status change and single history update.	FR-21	Medium	1
KS-7	As Kitchen Staff, I need similar products to be grouped into preparation batches, so I could work effectively.	Orders in the board are classified depending on product and variation and include notes of each customer.	FR-26	Low	2
KS-8	As Kitchen Staff, I need	Valid code of Ready order	FR-25	Medium	2

	the ability to scan a QR code of the customer upon delivery so collection is fast and traceable.	turns it into Collected state; invalid or used code will be rejected.			

Order Status Flow:
Confirmed-> Preparing->Ready->Collected. A cancelled order exits the active flow and keeps its history. No status may be skipped.